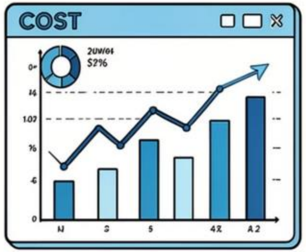
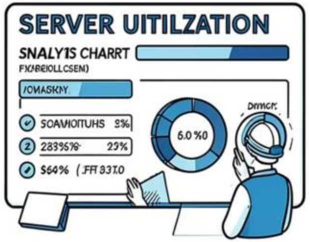
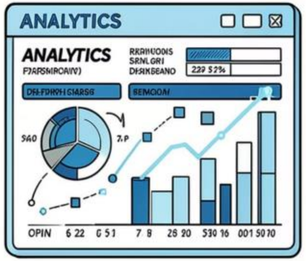
Optimization of Service Systems Using Queuing Theory and Game Theory

Research Project | Operations Research | Mathematical Modeling | Data Analytics





Why this Research Matters

Customer Waiting

Long waits undermine satisfaction and retention.

Cost Optimization

Balance staffing costs against service quality.

Resource Utilization

Improve efficiency across servers and counters.

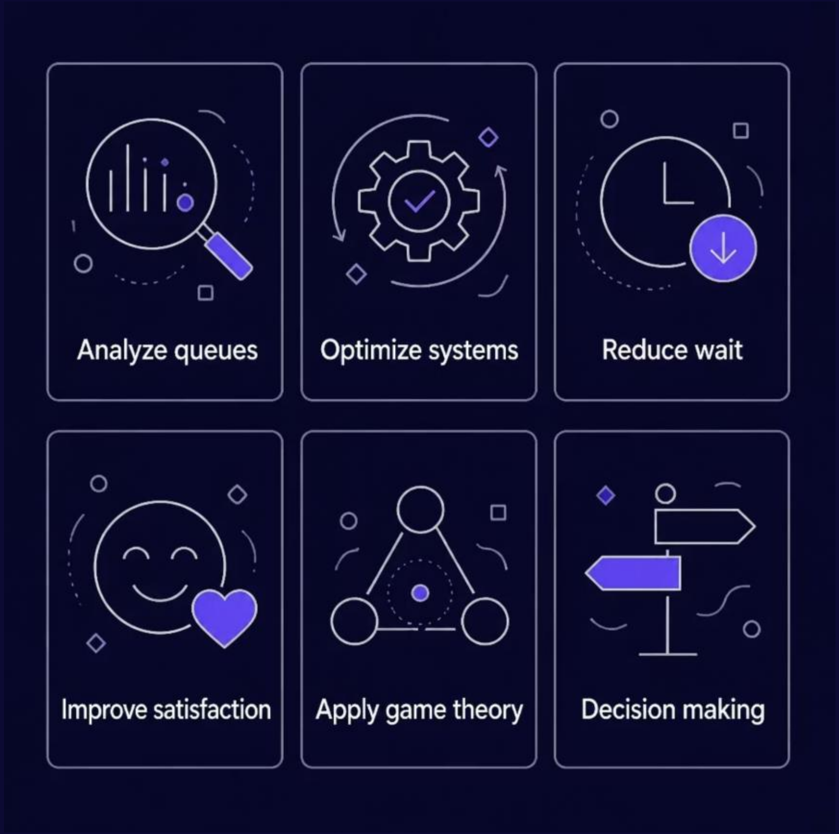
Data-Driven Decisions

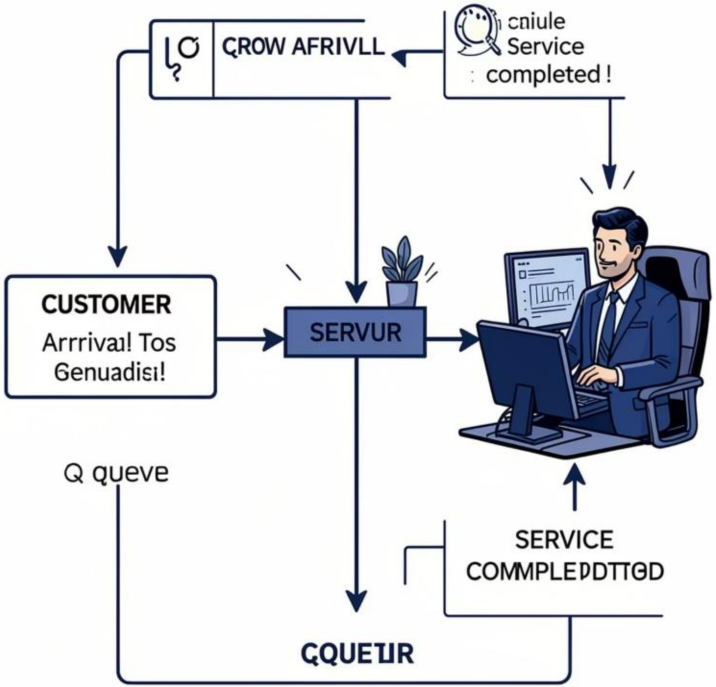
Analytics enable adaptive, real-time policies.

Research Objectives

Condensed project goals as an infographic.

- Analyze Queuing Models
- Optimize Service Systems
- Reduce Waiting Time
- Improve Customer Satisfaction
- Apply Game Theory
- Improve Decision Making
- Study Banking Applications





Introduction to Queuing Theory

Core parameters: Arrival rate (λ), Service rate (μ), Utilization ($\rho = \lambda / \mu$).

- **Key Metrics**

  Average queue length (L_q), average system size (L), waiting time (W_q), system time (W).

- **Little's Law**

  $L = \lambda W$ — universal balance for stable systems.

Introduction to Game Theory

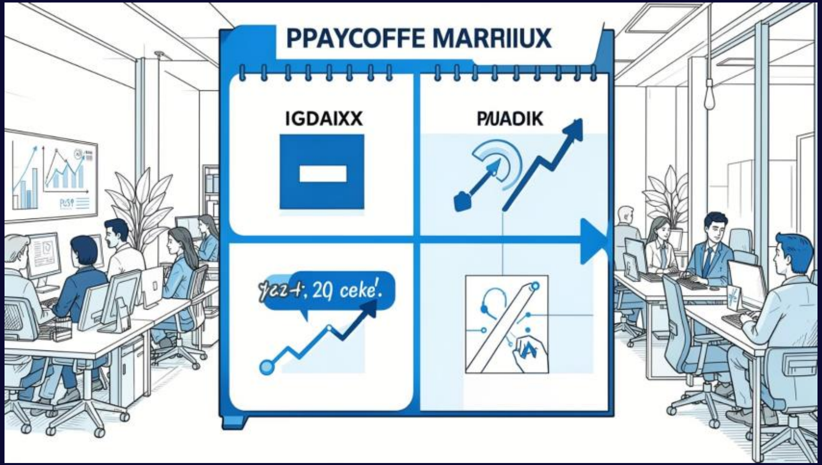
Strategic framework for actors who choose to optimize personal payoff.

Nash Equilibrium

No unilateral profitable deviation.

Competition & Pricing

Providers compete on price and capacity.



Applications: customer join/balk decisions, provider capacity & pricing games, mechanism design to align incentives.

Mathematical Models Employed

M/M/1

Markovian arrivals & service; single server. $\rho = \lambda/\mu$; $W = 1/(\mu-\lambda)$.

M/M/c

c servers, Erlang C for waiting probability; $Wq = C(c,\lambda/\mu)/(c\mu-\lambda)$.

M/G/1

General service distribution; Pollaczek–Khinchine: $Lq = \rho^2(1 + Cs^2)/[2(1-\rho)]$.

Real-Life Applications of Game Theory in Service Systems

Game theory provides a powerful framework for understanding and optimizing complex interactions within service environments, influencing both customer behavior and provider strategies.

Strategic Customer Behavior

Customers make rational decisions on joining, balking (not joining), or reneging (leaving the queue) based on perceived costs and benefits.

Provider Pricing & Capacity

Service providers strategically set prices and manage capacity to maximize profit, considering competitor actions and customer demand.

Queue Management & Incentives

Designing optimal priority queuing systems and congestion pricing mechanisms to balance efficiency with fairness.

Information Disclosure

The impact of transparently sharing wait times, service levels, or availability on customer choices and overall system performance.

Banking Case Study — Service Optimization

Problem

High waits at counters and ATMs during peaks.

Solutions

Queue modeling (M/M/c), priority lanes, dynamic staff scheduling.

Outcomes

Reduced Wq, better counter allocation, lower operating cost, improved CX.

Applications in Data & Decision Science

Predictive Analytics

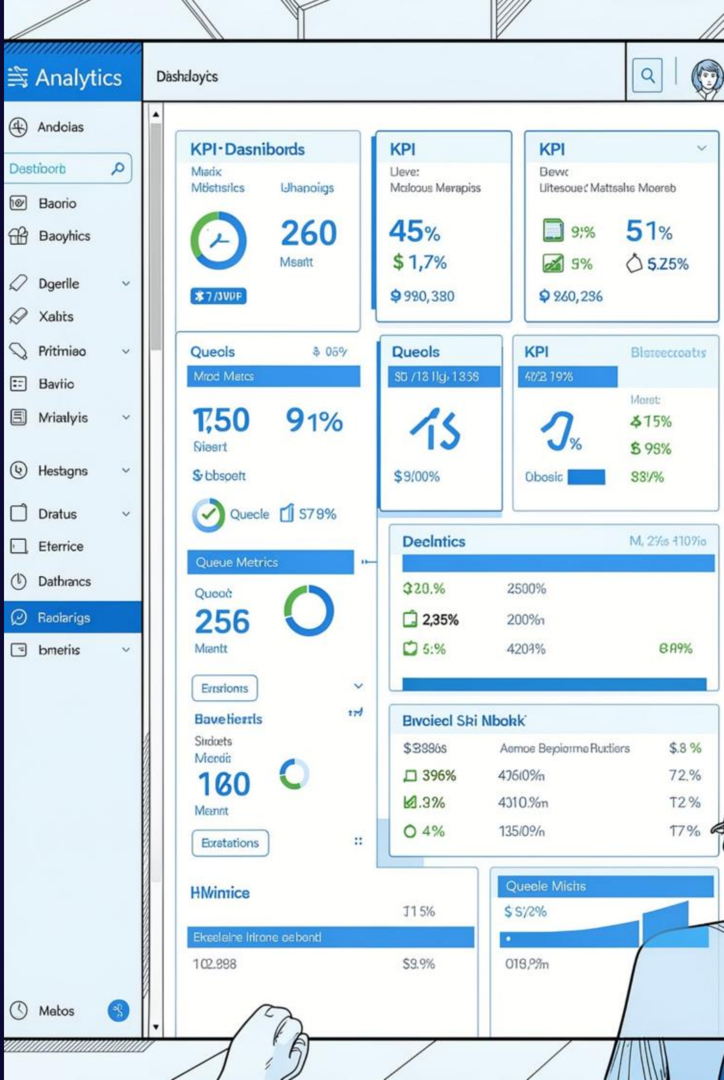
Forecast arrival patterns to staff proactively.

Optimization

Calibrate capacity and pricing to minimize total cost $TC = Cs \cdot c + Cw \cdot L$.

ML & Automation

AI agents making real-time routing and admission decisions; prospect for reinforcement learning games.



Key Takeaways, Future Scope & Call to Collaborate

Key Learnings

Queuing + Game Theory captures mechanics + behavior — essential to fix over-congestion and align incentives.

Skills & Methods

Mathematical modeling, optimization, mechanism design, data analytics, AI-assisted decision systems.

Future Directions

AI-driven agents, mean-field games for large systems, transient optimization for spikes, empirical validation.

Discover collaboration opportunities — researchers, analysts, and operations teams welcome. Contact for full report & datasets.

Thank You!

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M.Sc. Mathematics | Aspiring Data Analyst

"Transforming mathematical models into real-world solutions."

Research Interests

- Operations Research
- Queuing Theory
- Game Theory
- Mathematical Modeling
- Data Analytics

Let’s Connect

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